

Intra- and interspecific relations within *Laccaria bicolor sensu lato*

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Interstock pairing analyses were undertaken using isolates obtained from a number of specimens belonging to *L. bicolor sensu lato*. Three intersterility groups were detected among the tested North American isolates. The two employed Swedish isolates were completely intercompatible with one North American group but only somewhat intercompatible with the other two groups. These data are more or less concordant with those obtained by RFLP analysis of mtDNA and rDNA. Phenetic analyses of these and other collections revealed that the intersterility groups could be delimited on basidioma morphology. Synthesis of these data indicate that the North American members of the complex should be recognized as three distinct species: *L. bicolor sensu stricto*, *L. nobilis*, and *L. trichodermophora*. Data is needed on additional European collections referable to the *L. bicolor* complex to determine the significance of the observed anomalies in interstock pairing data and the divergence in rDNA of these isolates from North American isolates. These studies support the use of a multifaceted approach to solving taxonomic problems and obtaining information on speciation and the biology of agarics that form ectomycorrhizae.

The biological species concept has been used relatively rigorously for a number of years within the Aphyllophorales (e.g. Boidin, 1977; Boidin & Lanquetin, 1984a; Hallenberg, 1984). This paradigm also has been utilized increasingly with members of the Agaricales, especially saprophytic and coprophilous taxa (e.g. Lange, 1952; Vilgalys & Miller, 1983, 1987; Boidin, 1986; Fries, 1987; Jacobsson, 1989) and the Heterobasidiomycetes (e.g. Duncan & MacDonald, 1967; Boidin, 1986; Wells & Wong, 1989).

This paper presents a synthesis of the data obtained to date on *Laccaria bicolor* (Maire) Ort. *sensu lato* (Agaricales, Tricholomataceae). These data are part of a multifaceted study utilizing selected members of the genus *Laccaria* Berk. & Br. aimed at testing the utility of employing interstock pairing studies in conjunction with analyses of select macromolecules for solving systematic and biological problems in fungi that form ectomycorrhizae (Mueller, 1982, 1984, 1985; Fries & Mueller, 1984; Gardes *et al.*, 1990, 1991a,b). Currently it is unclear what the consequences of the differences in ecology and biology between ectomycorrhizal and saprobic fungi may have on long distance dispersal, gene exchange and speciation (Fries, 1987).

Laccaria bicolor sensu lato was chosen for these studies because it forms a conspicuous part of the mycota of the North Temperate zone with members of this complex often the most commonly encountered *Laccaria* in many areas (Mueller, 1982, 1990). Isolates of specimens referable to this group also have been used in a number of studies on

ectomycorrhizae (see Gardes *et al.*, 1990b). Finally, members of this complex have been prime candidates for studies examining the biology and speciation in agarics that form ectomycorrhizae because they can be manipulated in the laboratory relatively easily (e.g. Fries & Mueller, 1984; Kropp & Fortin, 1988; Armstrong, Fowles & Rygiewicz, 1989; Doudrick & Anderson, 1989; Gardes *et al.*, 1990, 1991a,b).

Various taxonomic interpretations of the observed variability in this complex have been proposed (see Singer, 1977; Mueller, 1985). In his dissertation, Mueller (1982) recognized four species in this complex: *L. bicolor*, *L. farinacea sensu* Singer non Hud., *L. nobilis* G. M. Mueller, and *L. trichodermophora* G. M. Mueller. Fries & Mueller (1984), using only Swedish isolates, uncovered no intersterility groups within the tested isolates which were referable to both *L. bicolor* and *L. farinacea sensu* Singer (Fries & Mueller, 1984). The other two taxa, *L. nobilis* and *L. trichodermophora*, are restricted to North America (N.A.), and were not included in that study (Fries & Mueller, 1984). Intersterile isolates were reported, however, from North American material of *L. bicolor sensu lato* by Kropp & Fortin (1988) and Doudrick & Anderson (1989) but with no indication as to which, if any, of these segregate taxa their isolates belonged.

The results of this study which used a number of collections and isolates referable to *L. bicolor sensu lato* obtained from N.A. and Europe support the recognition of three N.A. species within *L. bicolor sensu lato*: *L. bicolor sensu stricto*, *L. nobilis* and *L. trichodermophora*. *Laccaria farinacea sensu* Singer was shown

to be conspecific with *L. trichodermophora*. Questions remain, however, regarding the relationship between N. American and European populations of *L. bicolor sensu lato*.

MATERIALS AND METHODS

Procedures for obtaining collections and isolates

Specimens and cultures of *L. bicolor sensu lato* were collected from a wide variety of habitats throughout much of the continental U.S.A., southern Canada and central Sweden. Collections were made following standard procedures and deposited in F, TENN, UPS or WTU (Holmgren *et al.*, 1981). Monokaryotic and multispore cultures were obtained following the techniques of Fries (1983) and Fries & Mueller (1984). The resulting isolates were stored on N6:5 medium (Fries, 1983) at 2 °C. All isolates are housed in the mycological culture collection at Field Museum. Table 1 lists the 57 collections/isolates used in the following analyses.

Interstock pairing studies

Inocula for pairing tests were obtained from the margin of monosporous mycelia growing on Petri plates containing N6:5 medium. Isolates to be paired were placed approximately 10 mm apart on N6:5 plates and allowed to grow together (2–4 wk). After an additional 1 wk or more, plugs of tissue were cut from the interface and placed on fresh N6:5 plates. Mycelium growing from these plates were checked for the presence or absence of clamp connexions by examining them through the bottom of the inverted Petri plate at 200× magnification. Pairings that resulted in hyphae that bore clamp connexions were considered positive and those that did not yield clamped hyphae were scored as negative. Two or more testers for each stock (monokaryotic isolates originating from one basidioma), each containing different mating type alleles, were used, when possible. Only one isolate was used in the infrequent cases where a sufficient number of isolates could not be obtained to determine compatibility relationships between isolates in the stock.

Terminology to accurately reflect the biology observed during pairing studies has not yet been standardized (Boidin, 1986; Wells & Wong, 1989; Chase & Ullrich, 1990*a, b*). In this paper the terms non-compatible and intracompatible are restricted to intrastock pairings. We use the terms intersterile for intercollection pairings that do not form clamp connexions and intercompatible (rather than infertile) for positive intercollection matings when no data regarding fruiting and progeny analysis are available.

Phenetic analyses

Phenetic analyses were performed using NTSYS-pc version 1.40 (Rohlf, 1988). Characters and character states employed are listed in Table 2 and are a subset of the 30 characters used by Mueller (1985). After scoring each collection (OTU = Operational Taxonomic Unit) for morphological and cultural characters, the resulting data matrix was standardized and a similarity coefficient (Average Manhattan distances or Av-

erage taxonomic distance) calculated between each of the OTUs. The resulting matrix of coefficients was then analysed to reveal taxonomic structure using either a sequential technique, UPGMA (unweighted pair-group arithmetic average) cluster analysis or an ordination technique, PCA (principal components analysis).

RESULTS

Interstock pairing studies

Pairings among members of a stock confirmed previous reports that intraspecific compatibility was bifactorial in members of this complex (Fries, 1983; Fries & Mueller, 1984; Kropp & Fortin, 1988; Doudrick & Anderson, 1989). No false clamps or barrage zones, indicating common B or A matings, respectively, were observed. Clamp connexions were normally only found near the juncture of the two paired isolates indicating that nuclear migration was either very slow, or did not occur, on the media that we employed (i.e. N6:5).

Three intersterility groups were identified within the tested N.A. isolates belonging to *L. bicolor sensu lato* (Table 3). Each of the groups had a high level of intercompatibility among themselves but were almost completely intersterile with the other two groups. Confrontations among members of an intersterility group usually resulted in positive pairings with 78, 68 and 84% of the pairings yielding clamp connexions within 'group 1' (isolates 1744–2306), 'group 2' (isolates 1856–2118) and 'group 3' (isolates 1873–SAR9), respectively. Between-group intercompatibility was very low: group 1 × group 2 = 7.6%, group 1 × group 3 = 2% and group 2 × group 3 = 3.6%. Most of the observed positive pairings between groups occurred between some members of group 1, all of which were isolated from the southeastern U.S.A., and the isolates of group 2 originating from the Great Lakes region (Michigan and Ontario). Only three positive pairings (1781 × 2118, 2306 × 2027 and 1856 × 2061) were observed out of the over 120 cross-continental pairings made. Isolates in groups 1, 2 and 3 are referable on morphological characters to *L. farinacea sensu* Singer non Hud. (Singer, 1977), *L. bicolor sensu stricto*, and *L. nobilis*, respectively, except for isolate 2306 (group 1) which is referable to *L. trichodermophora* (Mueller, 1982, 1984, 1985). *Laccaria farinacea sensu* Singer and *L. bicolor sensu stricto* are equal to bi1 and bi2, respectively, in Mueller (1985).

The results of interstock pairings between N.A. Gulf Coast isolates referable to either *L. farinacea sensu* Singer (2282, 2283, 2308, 2321, 2372) or to *L. trichodermophora* (2306, 2315, 2319, 2324, 2325, 2329) are presented in Table 4. Almost all of the tested isolates were intercompatible with all of the other tested isolates. Isolates 2282 and 2306 are members of intersterility group 1 indicating that all of the tested Gulf Coast isolates belong to that group.

The results of interstock pairings between the N.A. isolates and two Swedish isolates are presented in Table 5. Two compatible isolates were employed for both of the Swedish testers. The Swedish testers were intercompatible with all members of N.A. intersterility group 2. Intercompatibility was reduced between the Swedish testers and the other two N.A. intersterility groups. Clamp connexions were formed in only

Table 1. Geographical origin and disposition of collections used in the phenetic analyses and pairing studies

ID no. ^{a, b}	Date	Country, State/Prov. County/Park	Herbarium
360	12 Sep. 82	SWED, Uppland, Uppsala	UPS
993	6 Sep. 80	U.S.A., TN, Sevier	TENN
994	6 Sep. 80	U.S.A., TN, Sevier	TENN
1014	17 Sep. 80	U.S.A., NC, Hendersonville	TENN
1024	19 Sep. 80	U.S.A., NC, Hendersonville	TENN
1060	5 Dec. 80	U.S.A., MS, Harrison	TENN
1062	5 Dec. 80	U.S.A., MS, Harrison	TENN
1067	5 Dec. 80	U.S.A., MS, Harrison	TENN
1071	6 Dec. 80	U.S.A., MS, Harrison	TENN
1225	26 Sep. 81	U.S.A., ID, Bonner	TENN
1230	26 Sep. 81	U.S.A., ID, Bonner	TENN
1264	3 Oct. 81	CAND, BC, Alice Lake Provincial Park	TENN
1293	13 Oct. 81	U.S.A., WA, King	TENN
1352	24 Oct. 81	U.S.A., WA, King	TENN
1353	24 Oct. 81	U.S.A., WA, King	TENN
1367	24 Oct. 81	U.S.A., WA, Kittitas	TENN
1372	24 Oct. 81	U.S.A., WA, Kittitas	TENN
1478	13 Nov. 81	U.S.A., OR, Tillamook	TENN
1591	3 Oct. 82	SWED, Bohuslan, north of Gothenburg	UPS
1744	16 Sep. 83	U.S.A., NC, Jackson	TENN
1771	19 Sep. 83	U.S.A., NC, Macon	TENN
1773	19 Sep. 83	U.S.A., NC, Macon	TENN
1777	20 Sep. 83	U.S.A., SC, Oconee	TENN
1781	20 Sep. 83	U.S.A., SC, Oconee	TENN
1804	22 Sep. 83	U.S.A., NC, Macon	F
1856	6 Sep. 84	U.S.A., MI, Cheboygan	F, WTU
1873	7 Sep. 84	U.S.A., MI, Mackinac	F, WTU
1881	10 Sep. 84	U.S.A., MI, Cheboygan	F, WTU
1892	13 Sep. 84	U.S.A., MI, Marquette	F, WTU
1931	18 Sep. 84	CAND, ONT, Algonquin Provincial Park	F, WTU
1932	18 Sep. 84	CAND, ONT, Algonquin Provincial Park	F, WTU
1934	18 Sep. 84	CAND, ONT, Algonquin Provincial Park	F, WTU
1981	6 Oct. 84	U.S.A., WA, Kittitas	F, WTU
1985	8 Oct. 84	U.S.A., WA, Kittitas	F, WTU
2008	15 Oct. 84	U.S.A., WA, Pierce	F, WTU
2013	18 Oct. 84	U.S.A., WA, Kittitas	F, WTU
2020	28 Oct. 84	U.S.A., WA, San Juan	F, WTU
2021	28 Oct. 84	U.S.A., WA, San Juan	F, WTU
2027	28 Oct. 84	U.S.A., WA, San Juan	F, WTU
2038	10 Oct. 84	U.S.A., WA, Gray's Harbor	F, WTU
2048	10 Oct. 84	U.S.A., WA, Gray's Harbor	F, WTU
2061	10 Oct. 84	U.S.A., WA, Gray's Harbor	F, WTU
2066	18 Nov. 84	U.S.A., CA, Mendocino	F, WTU
2086	22 Nov. 84	U.S.A., CA, Humboldt	F, WTU
2118	2 Dec. 84	U.S.A., CA, Mendocino	F, WTU
2282	26 Nov. 85	U.S.A., TX, Hardin	F
2283	26 Nov. 85	U.S.A., TX, Hardin	F
2306	29 Nov. 85	U.S.A., TX, Polk	F
2308	29 Nov. 85	U.S.A., TX, Polk	F
2315	29 Nov. 85	U.S.A., TX, Tyler	F
2319	29 Nov. 85	U.S.A., TX, Tyler	F
2321	29 Nov. 85	U.S.A., TX, Tyler	F
2324	29 Nov. 85	U.S.A., TX, Tyler	F
2325	29 Nov. 85	U.S.A., TX, Tyler	F
2329	1 Dec. 85	U.S.A., LA, St. Tammany	F
2372	6 Dec. 85	U.S.A., MS, Perry	F
SAR-9.1	9 Sep. 84	U.S.A., WA, Chelan	WTU

^a Collections housed at TENN are accessioned under the following TENN nos.: 993 (42521), 994 (42754), 1014 (42752), 1024 (42753), 1060 (42703), 1062 (42523), 1067 (42705), 1071 (42706), 1225 (42607), 1230 (42608), 1264 (42604), 1293 (42606), 1352 (42529), 1353 (42605), 1367 (42755), 1372 (42707), 1478 (42603), 1744 (44153), 1771 (44165), 1773 (44167), 1777 (44169), 1781 (44173).

^b Collections 360, 1981, 2008, 2086 and SAR-9.1 were supplied by N. Fries, M. Dilly, C. Scates and H. Barnhart, M. Seidle and S. Rehner, respectively.

Table 2. Characters and character states employed in the phenetic analyses

1. Lamellae colour	flesh colour = 1, violet = 2, vinaceous = 3
2. Basal mycelium colour	white = 1, violet (copious) = 2, violet (scant) = 3
3. Pileus texture	glabrous = 1, pruinose = 2, fibrillose = 3, fibrillose-scaly = 4, scaly = 5
4. Stipe striations	none = 1, slight = 2, moderate = 3, pronounced = 4
5. Pileus diameter	< 20 mm = 1, 20–45 mm = 2, > 46 mm = 3
6. Stipe length	< 40 mm = 1, 40–69 mm = 2, > 70 mm = 3
7. Stipe width	1–3 mm = 1, 4–7 mm = 2, > 10 mm = 3
8. Mean spore length	6.7–7.2 µm = 1, 7.3–7.7 µm = 2, 7.8–8.2 µm = 3, > 8.3 µm = 4
9. Mean spore width	5.7–6.2 µm = 1, 6.3–6.7 µm = 2, 6.8–7.2 µm = 3, 7.3–7.7 µm = 4, 7.8–8.2 µm = 5
10. Mean echinulae length	≤ 1 = 1, > 1 = 2
11. Pileipellis arrangement	interwoven = 1, fasciculate = 2, trichodermium = 3, individual perpendicular hyphae = 4
12. Pileipellis terminal hyphae	filamentous to clavate or barrel-shaped = 1, capitate = 2

21% of the confrontations between the Swedish isolates and group 1 and 57% of the confrontations between the Swedish testers and group 3 isolates.

Phenetic analyses

Both cluster analysis (Fig. 1) and PCA (Fig. 2) yielded complementary results. The phenogram presented in Fig. 1 has a cophenetic correlation coefficient of 0.76 and is based on

a Manhattan distance coefficient matrix. Caution had to be used when analyzing this phenogram because of the few characters that were available (Table 2). However, it was possible to compare trends observed in this phenogram to the data obtained via interstock pairing studies discussed above. The first cluster (1014–1781) contains all of the isolates, except 2306, included in intersterility group 1 plus those OTUs labelled bi1 used in the previous study (Mueller, 1985) with two of the Michigan group 2 isolates (1856, 1934)

Table 3. Results of intercollection pairing studies using North American isolates of *L. bicolor sensu lato*. Intercompatible pairings were identified by the formation of clamp connexions and are represented by +; intersterile pairings did not form clamp connexions and are represented by –. Each data point represents results from at least two replicate crosses and normally represents pairings made with two intracompatible isolates from each stock.

	1744	1771	1773	1777	1781	1804	2282	2306	1856	1882	1892	1931	1934	1985	2008	2013	2020	2021	2027	2038	2066	2118	1873	1932	1981	2048	2061	2086	SAR9
Group 1																													
1774	+	+	+	+	+	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1771		+			+	+	+							-	-	-	-	-					-		-		-		
1773					-	-	+	+	-	-	-	-											-						
1777					-		+	+	-	-	-	-	-				-	-				-			-	-	-	-	-
1781						+	+		+	+	-	-	+	-	-	-	-	-	-	-	-	+	-	-	-	-		-	-
1804							+	-	-	-	-	-		-	-	-		-			-	-	-	-	-	-	-	-	-
2282								+	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2306									-	-	-	-	-	-	-	-			+	-	-	-	-	+	-	-		-	-
Group 2																													
1856										+	-	+	+	+	-	-	+	-	+	-	+	+	-	-	-	-	+	-	-
1882											+	+	+	-	+	+	+	+	-	+	+	-	-	-	-	-		-	-
1892											-	+	-	-	-	-	-	-		+		+	-	-	-	-		-	-
1931												+	+	-	+	-	-	-	-			+	-	-	-	-	-	-	-
1934													-	-	-	+	-	+	+	+	+	-	-	-	-	-	-	-	-
1985														+	+	-	+	-	+	+	+	+	-	-	-	-	-	-	-
2008															+		+	+	+	+	+	+	-	-	-	-	-	-	-
2013																+	+	+	+	+	+	+	-	-	-	-	-	-	-
2020																	+	+	+	+	+	+	-	-	-	-		-	-
2021																				+	+	+			-	-		-	-
2027																					+	+	+		-	-	-	-	+
2038																						+	+	-	-	-	-	-	-
2066																							+	-	-	-	-	-	-
2118																									-	+	-	-	-
Group 3																													
1873																								+	+		+	+	+
1932																									+	-		+	+
1981																										+	-	+	+
2048																											+	+	+
2061																												-	+
2086																													+
SAR9																													

Table 4. Results of intercollection pairing studies using isolates from Gulf Coast (Louisiana, Mississippi and Texas) collections referable to *L. trichodermophora* and *L. farinacea sensu* Singer. Intercompatible pairings were identified by the formation of clamp connexions and are represented by +; intersterile pairings did not form clamp connexions and are represented by —. Each data point represents results from at least two replicate crosses.

	2282	2283	2306	2308	2315	2319	2321	2324	2325	2329	2372
2282		—	+	+	+	+	+	+	+	+	+
2283			+	+	+	+	+	+	+	+	+
2306				+	+	+	+	+	+	+	+
2308					+	+	+	+	+	+	—
2315						+	+	+	+	+	+
2319							+	+	+	+	+
2321								+	+	—	+
2324									+	+	+
2325										+	+
2329											+
2372											

clustered relatively closely. Most of intersterility group 2 along with those OTUs determined to be bi2 (Mueller, 1985) comprise three discrete clusters: 1367–1372, 1225–2027 and 1264–1230. OTUs 2020 and 2118, the remaining members of group 2, cluster loosely together with OTU 2306, which was labelled *L. trichodermophora* and belongs to intersterility group 1. Members of intersterility group 3 (OTUs 1981, 1873, 2048, 2086) clustered together with the one tested Swedish collection (OTU 1591). The bottom tight cluster (OTUs 1067–1060) is composed of OTUs treated as *L. trichodermophora* by Mueller (1985) but which were not included in the interstock pairing tests.

Most of the differences between the expected composition of clusters, based on the interstock pairing data, and the observed composition are due to relatively minor differences in morphology. For example, the cluster containing OTUs 1367–1372 are primarily separated from the other main cluster of members of intersterility group 2 plus bi2 (OTUs 1225–2027) on lamellae colour (flesh colour rather than vinaceous), stipe striations (slight versus moderate to pronounced) and having slightly shorter echinulae on their basidiospores. Members of the third cluster of group 2 (OTUs 1264–1230) have flesh colour lamellae as in OTUs 1367–1372, echinulae length similar to members of cluster 1225–2027 and longer and wider stipes than members of either of these clusters. Two collections from the Great Lakes area (OTUs 1856, 1934) are clustered between the large cluster comprised of members of intersterility group 1 and some members of group 2 due to small basidioma size and the scant violet mycelium at their stipe base (both characteristics of members of group 1) and vinaceous lamellae (a characteristic of many members of group 2). OTU 2306 (labelled *L. trichodermophora*) shares several distinct features (e.g. pileipellis composed of a trichodermium with capitate terminal hyphae) with the bottom tight cluster (OTUs 1067–1060) but differs by having vinaceous lamellae and copious violet basal mycelium.

For PCA analysis, 10 eigenvectors were extracted with the first three axes (Fig. 2) accounting for 60% of the total variation. Of this amount, roughly 44, 36 and 20% were accounted for by axes 1, 2 and 3, respectively. Basal mycelium was loaded very high on the first axis with basidiospore length, basidiospore width and cuticular arrangement loaded high on the second axis and the third axis based primarily on

pileus texture, basal mycelium, stipe length and lamellae colour.

Four groups of OTUs were separated from each other on the PCA axes (Fig. 2). Three of these groups (OTUs 1014–1781, 1981–2086, 1067–1060) were also identifiable on the phenogram (Fig. 1) and correspond to intersterility group 1 plus bi1 (Mueller, 1985), intersterility group 3 plus the one treated Swedish isolate (OTU 1591), and the three OTUs labelled *L. trichodermophora* by Mueller (1985). OTUs referable to intersterility group 2, along with the OTUs that were labelled as bi2 in Mueller (1985), formed a loosely organized group of OTUs reflecting the large amount of morphological variation observed in this group.

TAXONOMIC TREATMENT

The three recognized species of the *L. bicolor* complex can be distinguished from other North American members of *Laccaria* by the violet mycelium at the base of their stipes when fresh; their relatively small [mean dimensions usually less than $8(-8.5) \times 7 \mu\text{m}$ excluding ornamentation], subglobose to broadly ellipsoid, moderately echinulate basidiospores; and violet somatic culture mat on MMN and N6:5 media (Mueller, 1984, 1985). They are morphologically similar to each other, however, and it is occasionally difficult to correctly identify members of this complex without data from intercollection pairing tests, especially if no data on macromorphology based on fresh material are available. This problem is most noticeable in the Great Lakes area where collections morphologically intermediate between *L. bicolor sensu stricto* and *L. trichodermophora* are encountered.

No collections from the northeastern U.S.A. or Atlantic Coastal provinces of Canada were included in this study so it is not yet possible to determine the range of *L. trichodermophora* in eastern North America. Our assumption is that *L. trichodermophora* is restricted to the southeastern U.S.A., and possibly temperate Mexico, since it was not found within the tested isolates from the Great Lakes area. To date, no material referable to *L. bicolor sensu stricto* has been found in the southeastern U.S.A.

Isolates from collections labelled as *L. trichodermophora* were intercompatible with isolates from collections referable to *L. farinacea sensu* Singer (= bi1, Mueller, 1985) (Table 4). Along

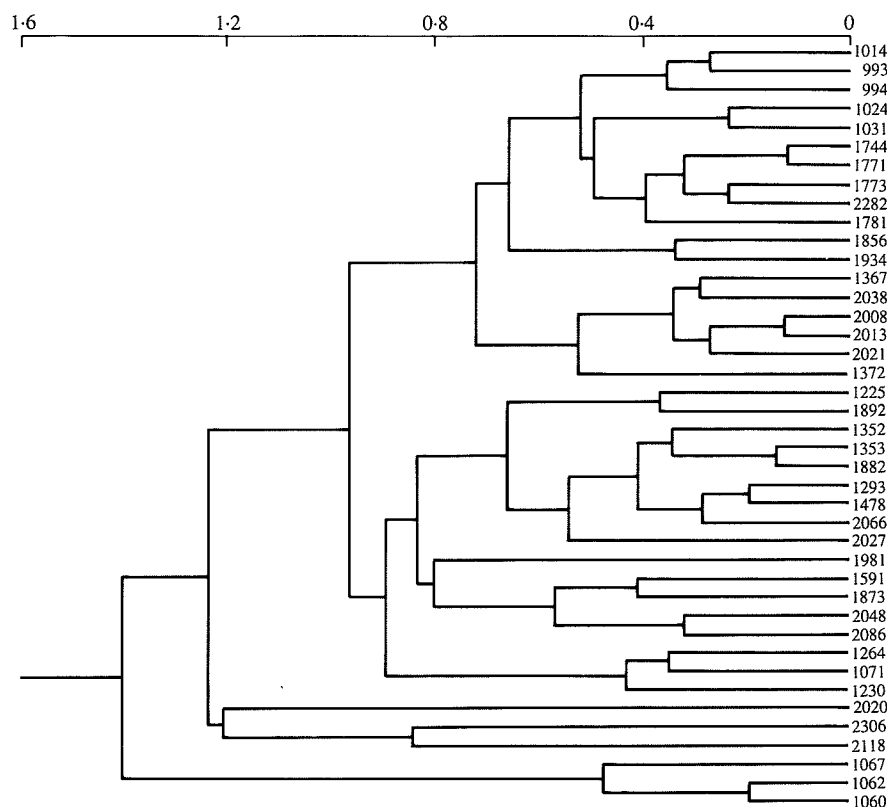


Fig. 1. Results of phenetic analyses performed on 41 OTUs of *L. bicolor sensu lato* using the 12 characters listed in Table 2. Phenogram produced by UPGMA cluster analysis and Manhattan distance coefficient matrix. Cophenetic correlation coefficient of this phenogram is 0.76.

with their restricted geographical range, all of these collections share the following features: small to moderate basidioma size; flesh colour lamellae (except OTU 2306); and scant, strongly hygrophanous violet mycelium at their stipe bases. The low phenetic similarity between members of this species indicated by the analyses (Figs 1, 2) and Mueller (1985) can be explained by the amount of variation observed in this group coupled with the fact that it was sometimes difficult to interpret certain characters such as basal mycelium colour (the violet pigmentation is strongly hygrophanous in this species), pileus texture and stipe striations due to age and weathering of the basidioma. As *L. trichodermophora* appears to be

restricted to North America, no previously published European name could be used for this taxon. As mentioned above, isolates from Swedish collections morphologically similar to this taxon were found to be intercompatible and contaxic with *L. bicolor sensu stricto* in Fries & Mueller's (1984) study of Swedish *Laccaria*.

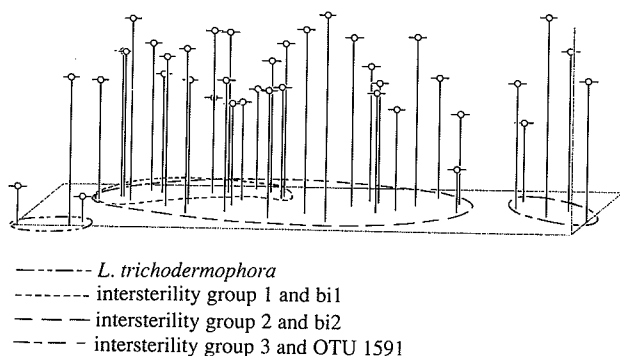


Fig. 2. Results of phenetic analyses performed on 41 OTUs of *L. bicolor sensu lato* using the 12 characters listed in Table 2. Three-dimensional graph of the first three axes produced by principal components analysis. These three axes explain 60% of the variation in the data matrix.

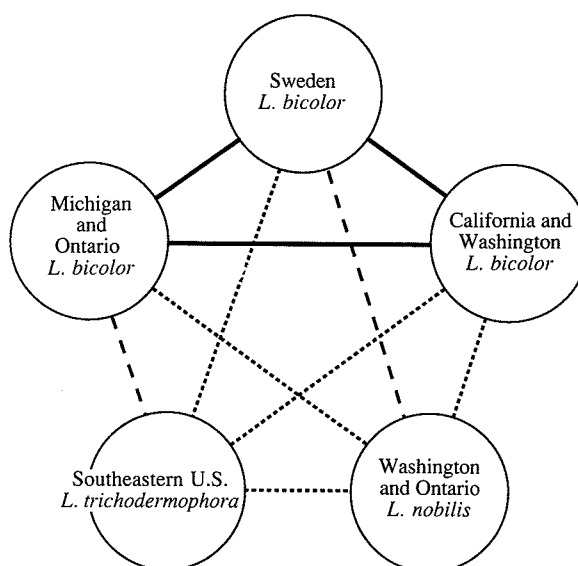


Fig. 3. Diagrammatic representation of intercollection pairing relationships between the tested populations of the recognized species. This figure is based on the results presented in Tables 3–5. Solid lines represent frequent matings, dashed lines represent occasional matings and dotted lines represent rare matings.

The concept of *L. nobilis* has been broadened from that originally published (Mueller, 1984). Interpretation of lamellae and basal mycelium colour was difficult in many of the collections included in the original description (Mueller, 1984) because they were very mature and weathered. The new collections (OTUs 1873, 1981, 2048, 2061, 2086) differed from the original description by having reddish brown basidioma, vinaceous lamellae and violet mycelium at the stipe base (as opposed to orange-brown basidioma, flesh colour lamellae and white basal mycelium). Since both the original

and the new collections all have large-sized, robust basidioma which become quite scaly in age, strongly striate to almost reticulate stipes, similar micromorphology and similar somatic culture mat morphology (Mueller, unpublished data), we are treating them as one species, *L. nobilis*. *Laccaria nobilis* has been identified among material collected throughout much of western North America (e.g. California, Colorado, Idaho, New Mexico, Oregon and Washington) and from the Great Lakes area (Michigan and Ontario) (Mueller, 1982, 1984).

Key to recognized species in the *Laccaria bicolor* complex

1. Pilei up to 80 mm broad, scaly; stipes robust (up to 10 mm broad at apex), strongly striate to reticulate, occasionally scaly *L. nobilis*
1. Pilei smaller, not scaly; stipes not robust, slightly to moderately striate, not reticulate or scaly. 2
2. Lamellae light vinaceous, fading to pinkish flesh colour when older; basal mycelium usually copious; basidioma usually pinkish to reddish flesh colour *L. bicolor*
2. Lamellae flesh colour to pinkish flesh colour when fresh; basal mycelium usually scant; basidioma usually orange-brown; south-eastern U.S.A. *L. trichodermophora*

DISCUSSION

Results of interstock pairings among the tested N.A. isolates support the recognition of several segregate taxa within the *L. bicolor* complex (Tables 3, 4). The observed frequencies of positive matings within and between intersterility groups conform, for the most part, to frequencies reported for taxa within other genera which are presumed to be recently diverged (e.g. Macrae, 1967; Anderson & Ullrich, 1979; Anderson, Korhonen & Ullrich, 1980; Boidin, 1986). The major source of noise in this data set was interstock pairings involving the isolates from the Great Lakes area (Michigan and Ontario) belonging to group 2 (1856, 1882, 1892, 1931, 1934). Seven of the ten matings between members of different intersterility groups involved these isolates. The relatively high incidence of negative pairings within group 2 (only 68% of the pairings between members of this group formed clamp connexions) also is attributable to these isolates.

These data indicate the probable recency of the speciation events within this complex (Boidin, 1986; Fries, 1987). Whether these matings between intersterility groups indicate that divergence between the intersterility groups is not yet complete, or are due to the Great Lakes area population of intersterility group 2 undergoing further speciation is not possible to predict until a usable phylogeny for these taxa is available. In any case, since many of the pairings involved more than one intracompatible isolate of a stock, the observed negative pairings within and between intersterility groups cannot be due solely to homozygosity of mating type alleles. The existence of a large number of mating factors occurring within each of the intersterility groups is supported by the findings of Doudrick & Armstrong (1989) that 10 A factors and 8 B factors existed within the 5 northern Minnesota *L. bicolor sensu lato* isolates that they tested. These data, therefore, strongly support the contention that there are multiple sterility genes acting in consort with the compatibility genes to control intercollection matings (Burnett, 1983; Wells & Wong, 1989; Chase & Ullrich, 1990a, b). It must also be remembered that it is impossible to tell from these studies if

the 'hybrid' pairings are fertile or even if geographical and/or ecological barriers would permit such pairings to occur in the field. Currently it is not possible to determine if a pairing is truly fertile (i.e. gives rise to viable progeny) in *Laccaria* and most other genera with species that form ectomycorrhizae because one can not routinely fruit the resulting dikaryons and test their basidiospores for viability. Thus it is impossible to tell what barriers to successful karyogamy and subsequent meiosis may be present in a dikaryon formed by pairing two isolates in the laboratory.

Phenetic analyses of the collections utilized above, along with those used in a previous study (Mueller, 1985), were undertaken to determine if the identified intersterility groups correlated with basidioma macro- and micromorphology. Differences observed in the growth rate and culture mat colour between the monokaryons and reconstituted dikaryons of tested stocks precluded using somatic culture mat data in these phenetic studies because many of the OTUs did not have a corresponding dikaryotic isolate of tissue culture origin. Attempts at obtaining tissue cultures, in addition to monokaryotic isolates, need to be made to enable use of somatic culture mat data in numerical analyses, both phenetic and cladistic. Culture mat data have proven to have significant taxonomic value in *Laccaria* (Mueller, 1985).

The resulting phenogram (Fig. 1) and three dimensional PCA graph (Fig. 2) indicate a relatively close correlation between intersterility groups and morphology with most of the members of each intersterility group being clustered in proximity to others belonging to the same intersterility group. Members of the large intersterility group 2 were the most dispersed on both diagrams with some of its members clustering more closely with members of group 1 or group 3 than others of their own intersterility group. This can be explained in part by the fact that the morphological variation inherent in this group may occasionally enable small morphological differences to overly influence the placement of certain OTUs on the phenogram (see RESULTS).

Because it is not possible to routinely fruit *Laccaria* in the laboratory, and consequently impossible to examine progeny,

we wished to obtain additional information to better interpret the biological and taxonomic implications of the observed interstock pairing behavior. Restriction fragment length polymorphisms (RFLPs) analysis of mitochondrial DNA (mt DNA) and nuclear ribosomal DNA (rDNA) were used as indicators of genetic similarities among *Laccaria* isolates (Gardes *et al.*, 1990, 1991a).

Based on the analysis of RFLP in the nuclear ribosomal DNA, all *L. bicolor* isolates were clearly distinct from the tested *L. laccata* Cooke, *L. proxima* (Boud.) Pat. and *L. amethystina* Cooke isolates (Gardes *et al.*, 1990). Although variability was observed (most probably in the non-coding regions) among N.A. *L. bicolor* isolates, they formed a relatively homogenous group which was, however, distinct from that of the two tested Swedish *L. bicolor* isolates. Furthermore, partial nucleotide sequences of the internal transcribed spacer (ITS) of the nuclear ribosomal repeat unit have shown that the degree of variation among three *L. bicolor* isolates (from two different biological species) was 1–2%, around 3% between *L. bicolor* and one tested *L. laccata* isolate and up to 5% between *L. bicolor* and the one *L. proxima* isolate tested (Gardes *et al.*, 1991a). All of these results suggest that genetic isolation between populations of *L. bicolor sensu lato* could have occurred recently with respect to the other tested species.

Much mtDNA variability was observed among the *L. bicolor* isolates with more than one mitochondrial phenotype observed within each of the *L. bicolor* intersterility groups (Gardes *et al.*, 1991a). However, most of the members of each intersterility group were usually clustered together on the phenograms presented by Gardes *et al.* (1990b, submitted) and no or few similarities were noticed between isolates of two different intersterility groups. For example, five isolates from group 2 (1293, 1367, 1478, 2008, 2021) were indistinguishable using three different mtDNA probes combined with two different restriction enzymes. They also clustered at more than 60% similarity with two other isolates of this group (1264, 2013) using *Bgl* II digests. They showed less than 10% similarity, however, with members of intersterility groups 1 and 3 using the same enzymes. The employed isolates of intersterility group 1 were usually clustered together but less tightly than the previous isolates on the phenograms of Gardes *et al.* (1991a). Members of intersterility group 3 (1873, 1981, 2061) were separated on the phenogram based on *Bam*H I digest but clustered together on the phenograms based on the two other employed digests (i.e. *Bgl* II and *Hind* III). Some exceptions did exist, however, such as isolates 1225 and 1014 (intersterility groups 2 and 1, respectively) which never clustered with members of their corresponding intersterility groups. These exceptions, however, do not greatly detract from the overall concordance of groupings of isolates based on mtDNA RFLPs and the observed intersterility groups.

Although three intersterility groups within the N.A. population of this complex were well delimited (Table 3), our results from intercontinental pairings (Table 5) do not fit any known pattern. A number of workers have reported full intercompatibility between European and N.A. populations of the same species (e.g. Hallenberg, 1984; Ginns, 1985; Boidin,

Table 5. Results of intercollection pairing studies using isolates from Sweden and the North American isolates used in Table 3. Intercompatible pairings were identified by the formation of clamp connexions and are represented by +; intersterile pairings did not form clamp connexions and are represented by –. Two intracompatible isolates were used for each of the two Swedish testers.

	360	1591
360		+
1591	+	
1744	–	–
1771		
1773	–	–
1777	–	–
1781	+	–
1804	–	–
2282	+	+
2306	–	–
1856	+	+
1882	+	+
1892	+	+
1931	+	+
1934	+	+
1985	+	+
2008	+	+
2013	+	+
2020	+	+
2021	+	+
2027	+	+
2038	+	+
2066	+	+
2118	+	+
1873	–	–
1932	–	–
1981	+	–
2048	+	+
2061	+	+
2086	+	+
SAR9	+	–

1986). There also have been several reports of one, two or more intersterile populations on one continent being partially intercompatible with a population on another continent (e.g. Mounce & Macrae, 1938; Macrae, 1967; David, Tortic & Jelic, 1974; Anderson *et al.*, 1980; Boidin & Lanquetin, 1984a, b; Chase & Ullrich 1990a, b). Boidin (1986) explains the reported partial intercompatibility between allopatric populations as being the result of geographical isolation precluding the necessity of the formation of a genetic isolation mechanism. This explanation has been used to explain similar situations in plants and animals (e.g. Grant, 1981).

In our study, the tested members of the Swedish population are completely intercompatible with one N.A. biological species but only partially intercompatible with the other two intersterility groups. Two plausible explanations can be given for the intercontinental relationships: (1) the Swedish members of the complex are contaxic and interbreeding with intersterility group 2; or (2) Swedish isolates and N.A. group 2 are geographically isolated, nonbreeding, separate biological species. Neither of these two hypotheses convincingly explain the observed intercontinental pairing data; e.g. 21 and 57% of the pairings between the Swedish isolates and isolates of N.A. intersterility groups 1 and 3, respectively, were positive (Table 5) even though members of groups 1 and 3 are almost

completely intersterile with the N.A. members of group 2.

Our data on RFLP analysis of mtDNA and rDNA do not resolve this issue. Gardes *et al.* (1990) reported significant divergence between the tested N.A. and Swedish isolates of *L. bicolor sensu lato* based on RFLP analysis of rDNA but little evidence of divergence within N.A. populations. Analyses of mtDNA, however, uncovered divergences within N.A. populations but no obvious divergences between N.A. and Swedish populations of *L. bicolor sensu lato* (Gardes *et al.*, 1991a). With data on only two European isolates, it is not possible to compare the amount of heterogeneity of either rDNA or mtDNA within the N.A. and European populations. Therefore a decision on whether or not members of the Swedish population and N.A. intersterility group 2 are contaxic has to be deferred.

In summary, interstock pairing tests support the recognition of three species within the North American population of *L. bicolor sensu lato*. Phenetic analyses have shown that these species can be delimited on morphological grounds. RFLP analyses at mtDNA indicate that these species are genetically distinct even though there was a significant amount of observed intraspecific variation. It is currently impossible, however, to determine whether the observed incompatibility between the two tested members of the Swedish population and N.A. intersterility group 2 indicate that the two populations actually exchange genetic information or are genetically isolated due to geographic isolation which precluded the development of a genetic isolation mechanism. For the present, we have treated members of these two populations (Sweden and group 2) as members of the same taxonomic species, *L. bicolor sensu stricto*. Members of intersterility group 1 belong to *L. trichodermophora* and members of group are referable to *L. nobilis*. Isolates from collections referable to *L. trichodermophora* and *L. farinacea sensu* Singer non Hud. were intercompatible and are treated as contaxic. Fig. 3 illustrates the taxonomic and intercompatibility relationships between the populations studied.

Phylogenetic relationships among these species are not yet determined. Because of the paucity of usable morphological characters, it will be necessary to utilize either extensive restriction site mapping or sequencing of selected macromolecules to develop a robust hypothesis of phylogenetic relationships within this group (e.g. Bruns & Palmer, 1989; Bruns *et al.*, 1989; Taylor & Natvig, 1989). This study, however, reemphasizes the necessity of having data on the breeding biology of a group to determine species composition and potential gene exchange between populations for understanding the biology, speciation and evolution in higher fungi, including genera with species that form ectomycorrhizae.

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